How Did Discrete Diffusion Become a Large Language Model?

From SEDD to LLaDA

발표일: 2026.10.12.
발표자: 이혜은

From SEDD to LLaDA

From score estimation to masked token prediction

SEDD (2024)
Discrete Diffusion

→

LLaDA (2025)
Large Language Diffusion Model

Enabled score matching in discrete diffusion models
Introduced **score entropy** as a training objective

Scaled masked diffusion to large language modeling.
Used weighted masked-token cross-entropy for both pretraining and instruction tuning.

$\nabla_x \log p_t(x)$ → $\frac{p_t(y)}{p_t(x)}$ → $p_\theta(x_0^i | x_t)$ → Weighted CE

Continuos diffusion | SEDD | RADD/MDLM | LLaDA

AR Already Works. Why Diffusion?

- AR works remarkably well
- Why do we need another generative formulation?

A Proven Success
- Strong language understanding and generation
- Scalable training and reliable performance
- Foundation of modern LLMs

A Fixed Generation Order
- Left-to-right token generation
- Each token depends on previous tokens

Is left-to-right generation necessary for a powerful language model?

Is Autoregression Necessary?

Generative Modeling
The Goal
Learn the probability distribution of language.

$$p_\theta(x) \approx p_{\text{data}}(x)$$

$$\underbrace{\max_\theta \mathbb{E}_{p_{\text{data}}(x)} \log p_\theta(x) \Leftrightarrow \min_\theta \mathrm{KL}(p_{\text{data}}(x) || p_\theta(x))}_{\text{Generative modeling principles}} .$$

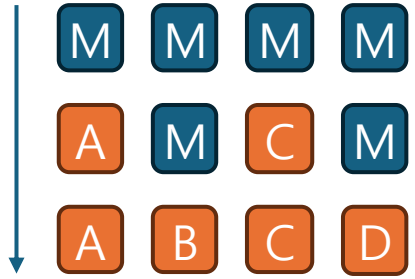
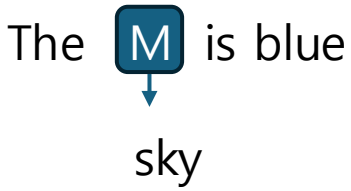
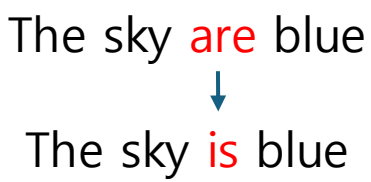
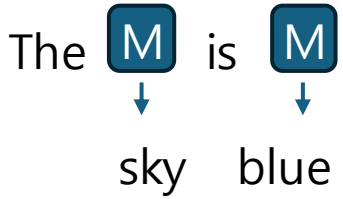
Autoregressive Modeling
One Way to Achieve It
Factorize the joint distribution into sequential conditional probabilities.

$$\underbrace{p_\theta(x) = p_\theta(x^1) \prod_{i=2}^{L} p_\theta(x^i \mid x^1, \ldots, x^{i-1}),}_{\text{Autoregressive formulation}}$$

Autoregression is one way to model language, not the only way.

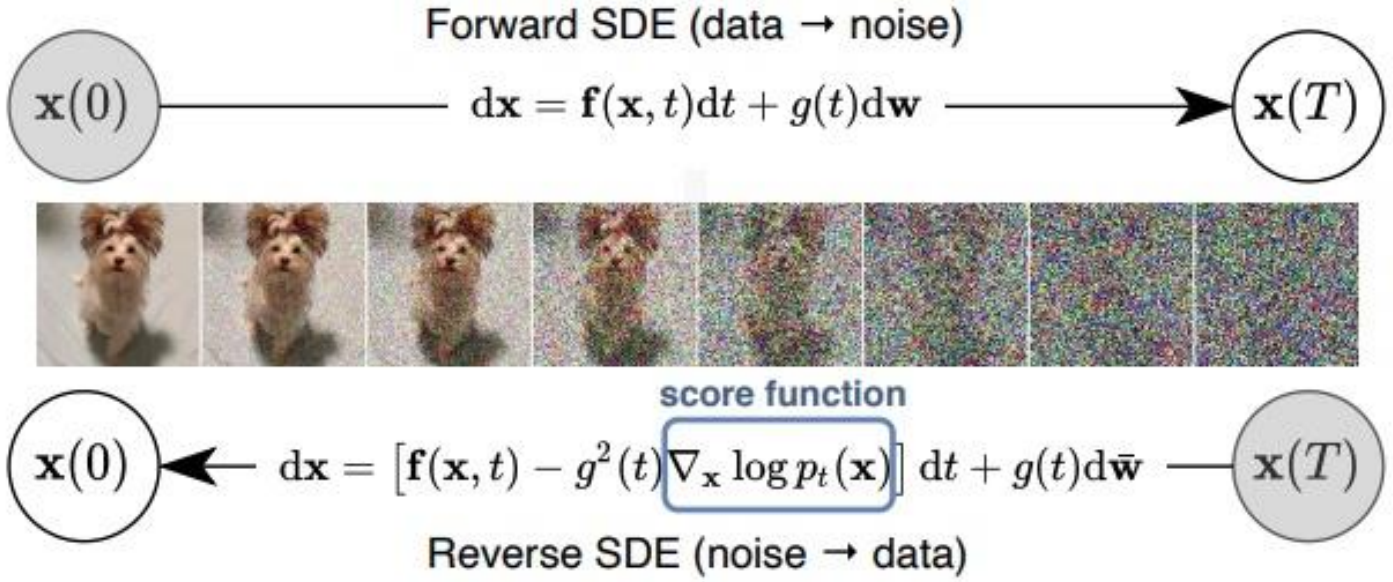
What We Want Beyond AR

- What Could Diffusion Offer?

Flexible Generation Order 	Bidirectional Context 
Iterative Refinement 	Parallel Token Prediction 

But Diffusion Was Born Continuous

- **Image Diffusion**



- **The Challenge**

  Images are continuous. Language is discrete.

  How can we define noise and denoising for tokens?

SEDD: Bridging Scores to Discrete Space

- Replace continuous gradients with probability ratios.

Continuous Diffusion
Score = Direction of Increasing Probability $\nabla_x \log p_t(x)$

How does probability change as x moves?

Discrete Diffusion (SEDD)
Concrete Score = Probability Ratio

$$s_\theta(x,t) \approx \left[\frac{p_t(y)}{p_t(x)}\right]_{y \neq x}$$

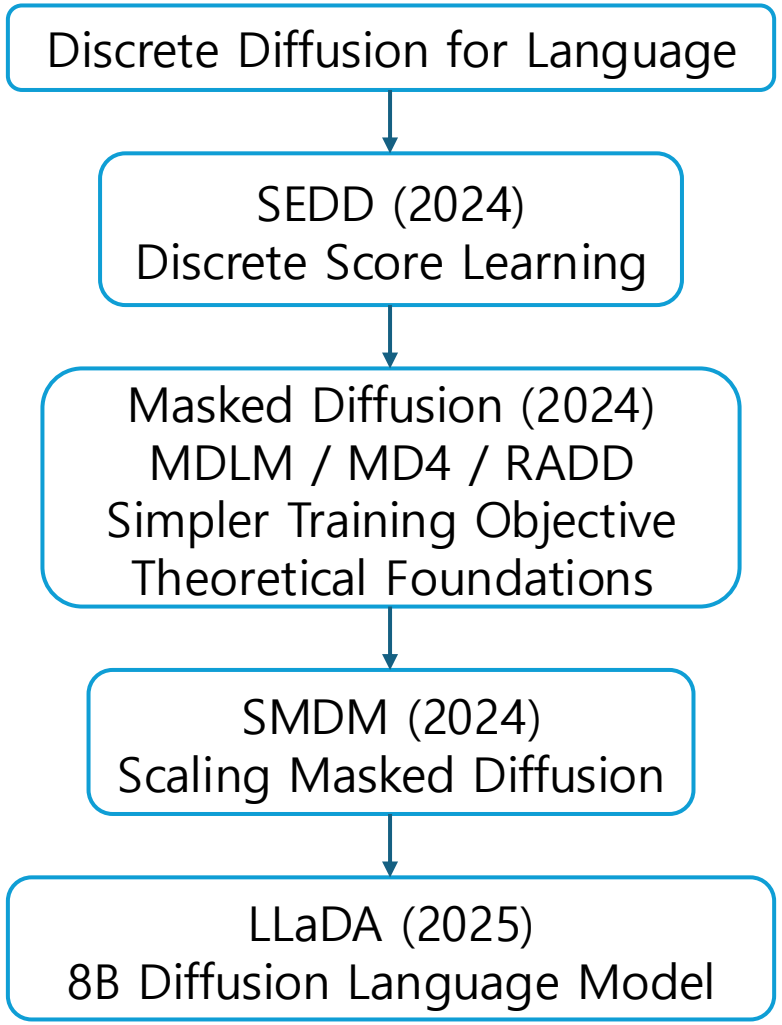
Current state (x): The [M] is blue
Candidate state (y): The sky is blue

How likely is state y relative to state x?

SEDD replaces continuous score estimation with discrete probability-ratio estimation.

A Two-Year Simplification

Conceptual evolution



- Discrete diffusion evolved from learning probability ratios toward simpler masked-token prediction and scalable language modeling.

- **SEDD — Learning Discrete Scores**

  How can we learn probability ratios to reverse discrete diffusion?

- **MDLM / MD4 / RADD — Simplifying Masked Diffusion**

  How can masking simplify the training objective and connect scores to token prediction?

- **SMDM / LLaDA — Scaling to Language Models**

  Can masked diffusion scale to competitive large language models?

- The key shift: From discrete score modeling to simplified masked diffusion—and finally, scaling to LLMs.

  https://github.com/ML-GSAI/LLaDA/blob/main/README.md

A Two-Year Simplification

7. Our journey in developing LLaDA?

LLaDA is built upon our two prior works, RADD and SMDM.

RADD demonstrated that the **training objective of LLaDA serves as an upper bound on the negative log-likelihood** of the model's distribution, a conclusion also supported by MD4 and MDLM. Furthermore, RADD was the first to theoretically prove that **masked diffusion models do not require time t as an input to Transformer**. This insight provides the theoretical justification for LLaDA's unmodified use of the Transformer architecture. Lastly, RADD showed that **the training objective of masked diffusion models is equivalent to that of any-order autoregressive models**, offering valuable insights into how masked diffusion models can overcome the reversal curse.

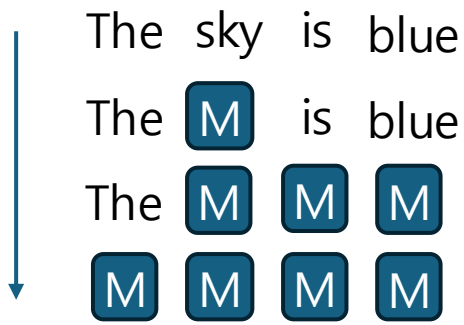
SMDM introduces the first **scaling law** for masked diffusion models and demonstrates that, with the same model size and training data, masked diffusion models can achieve downstream benchmark results on par with those of autoregressive models. Additionally, SMDM presents a simple, **unsupervised classifier-free guidance** method that greatly improves downstream benchmark performance, which has been adopted by LLaDA.

- RADD established the theory. SMDM demonstrated scalability. LLaDA brought them together.

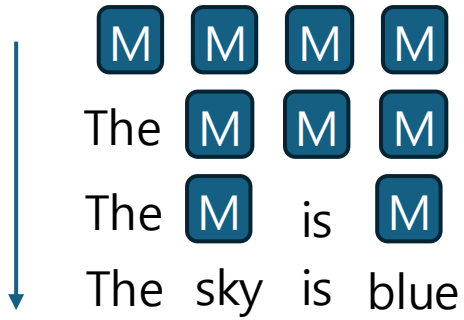
The Key Simplification: Just Mask It

- Instead of arbitrary token replacements, masked diffusion uses a single absorbing state: [MASK].

Forward Process – corrupt by masking



Reverse Process – generate by unmasking



Simple Corruption

Tokens are either preserved or masked.

Natural Training Target

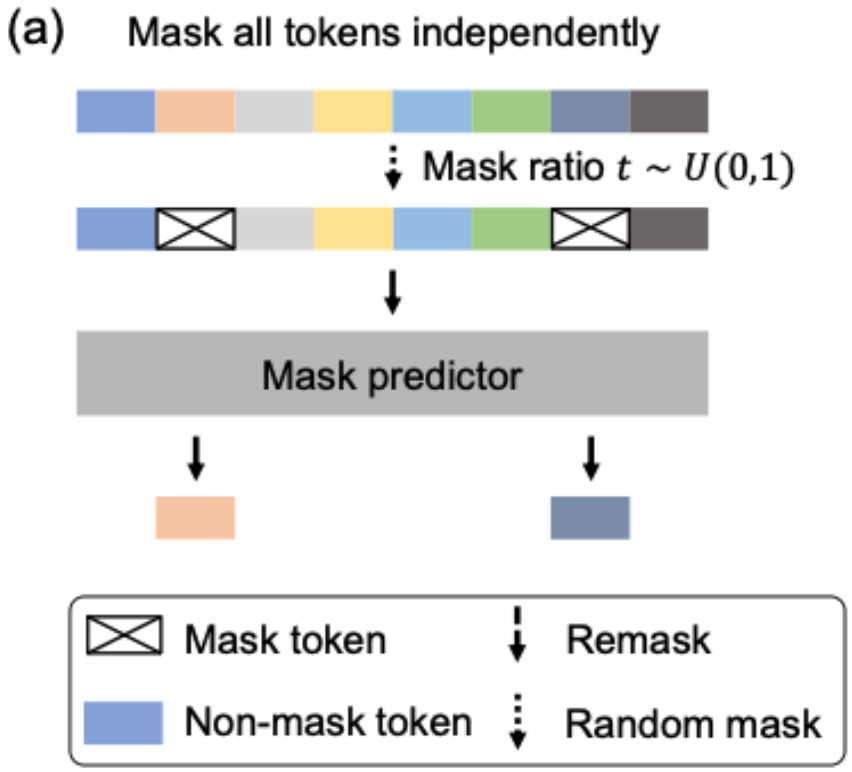
The original token is known at every masked position.

Flexible Generation

Masked positions can be predicted using bidirectional context.

Wait… Isn't This BERT?

LLaDA



	BERT	LLaDA
input	Randomly masked text	Randomly masked text
Masking ratio	Typically 15% selected	Varies across $t \in U[0,1]$
Training objective	Masked token CE	Weighted masked-token CE
Modeling goal	Contextual representations	Generative modeling of $p(x)$

Same CE, Different Meaning

$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t,x_0,x_t}\left[\frac{1}{t}\sum_{i=1}^{L}\mathbf{1}[x_t^i = \mathrm{M}]\log p_\theta(x_0^i|x_t)\right]$$

x_t : input(partially masked)

x_0 : training sample

t~U[0,1]

Theoretical foundation: MDLM / MD4 / RADD

$$\mathrm{P}(\text{unmask in } \Delta t|\text{masked at t}) = \frac{\Delta t}{t} \longrightarrow \sum_{\text{steps}} \frac{\Delta t}{t}\,\mathrm{CE}_{\text{masked}} \rightarrow \int_0^1 \frac{1}{t}\,\mathbb{E}[\mathrm{CE}_{\text{masked}}|t]dt$$

RADD : $-\log p_\theta(x_0) \leq \mathcal{L}_{\mathrm{LLaDA}}$

- The difference is not the CE itself, but how masked prediction is connected to sequence-level generative modeling.
- LLaDA learns the entire reverse diffusion process by training on randomly sampled masking levels, without explicitly simulating every reverse step.

LLaDA's Core Claim

- Can masked diffusion become a competitive large language model?

Learning the Data Distribution

$p_\theta(x) \approx p_{data}(x)$

Learn a sequence distribution

Likelihood-bound objective

Scaling to Large Language Models

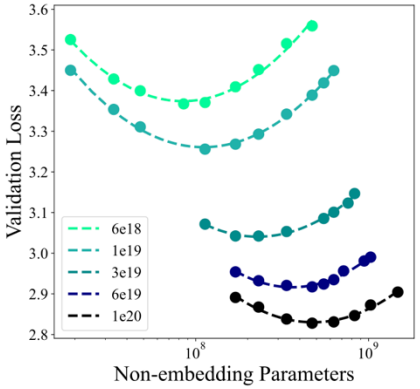
Masked Diffusion → 8B LLM

Scale parameters and training data

A principled generative model is not enough—it must also scale.

Scaling Masked Diffusion to 8B

SMDM



(b) MDMs.

- Scaling a diffusion language model with a familiar Transformer architecture.
- Transformer without Causal Mask

Table 5: **Model Architecture.** We report the architectural configurations for our 1B and 7B ARM baselines, the 1B and 8B LLaDA models, and the 8B LLaMA3 model.

	Our ARM Baseline 1B	LLaDA 1B	Our ARM Baseline 7B	LLaDA 8B	LLaMA3 8B
Layers	22	22	28	32	32
Model dimension	2048	2048	4096	4096	4096
Attention heads	32	32	32	32	32
Vocabulary size	126,464	126,464	126,464	126,464	128,000
FFN dimension	5634	5634	13,440	12,288	14,336
Key/Value heads	4	4	8	32	8
Total parameters	1.49 B	1.49 B	6.83 B	8.02 B	8.03 B
Non-embedding parameters	0.97 B	0.97 B	5.80 B	6.98 B	6.98 B

How LLaDA Is Trained

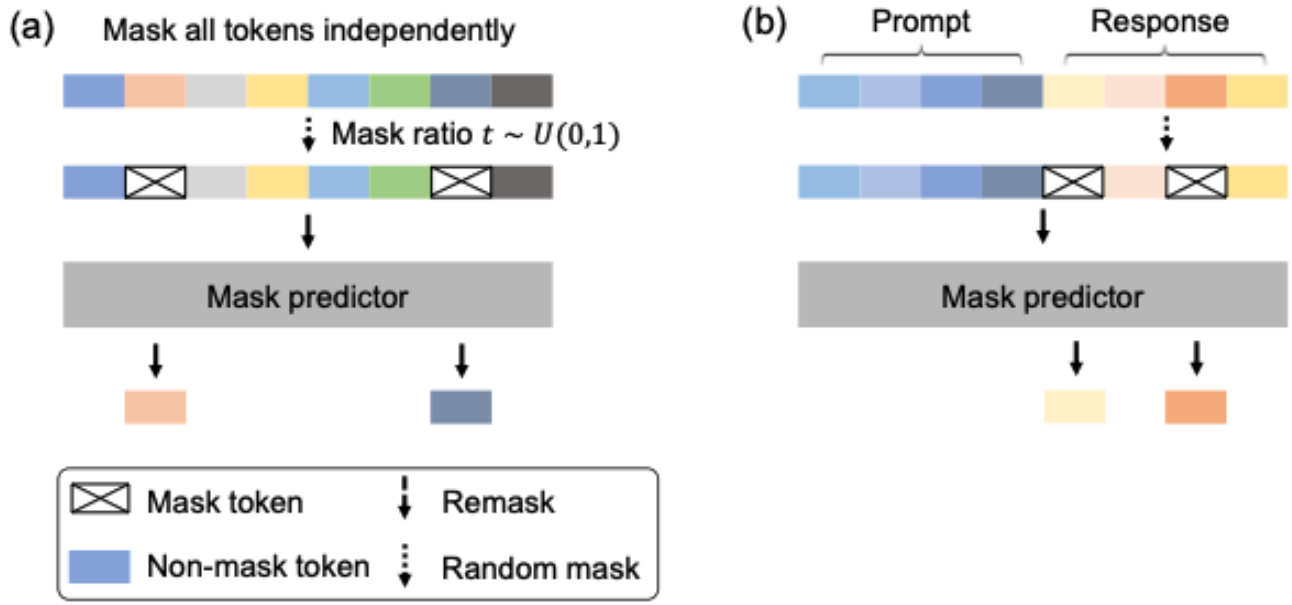
RADD :

$$p(x_0|x_t, t) = p(x_0|x_t)$$

No explicit timestep conditioning is required

Stage 1 : pretraining

Stage 2 : SFT

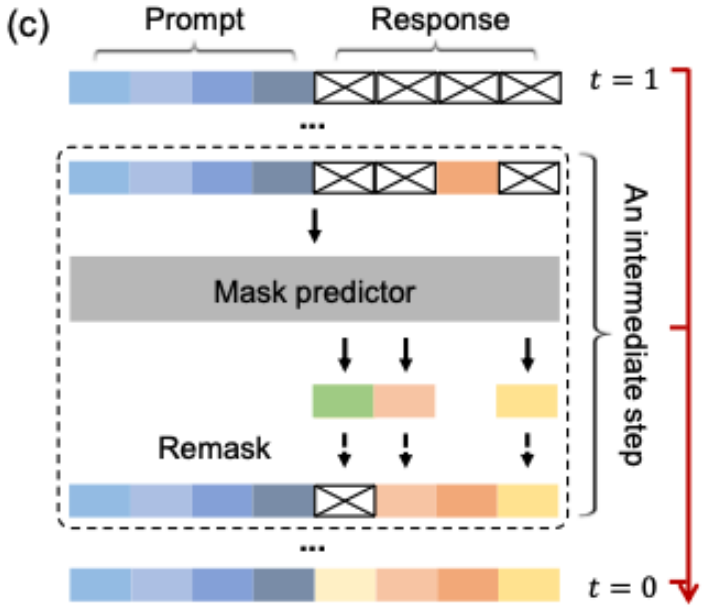


$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t,x_0,x_t}\left[\frac{1}{t}\sum_{i=1}^{L}\mathbf{1}[x_t^i = \mathrm{M}]\log p_\theta(x_0^i|x_t)\right]$$

$$-\mathbb{E}_{t,p_0,r_0,r_t}\left[\frac{1}{t}\sum_{i=1}^{L'}\mathbf{1}[r_t^i = \mathrm{M}]\log p_\theta(r_0^i|p_0, r_t)\right]$$

Same masked diffusion objective, different conditioning and masking scope.

How LLaDA Generates



Initialize with All Masks

Predict All Masked Tokens
Parallel prediction

Select tokens to reveal

Random remasking / low-confidence remasking

- Iterative denoising instead of next-token prediction
- Multiple tokens can be predicted in parallel, while the sequence is refined over multiple steps.

What Did LLaDA Achieve?

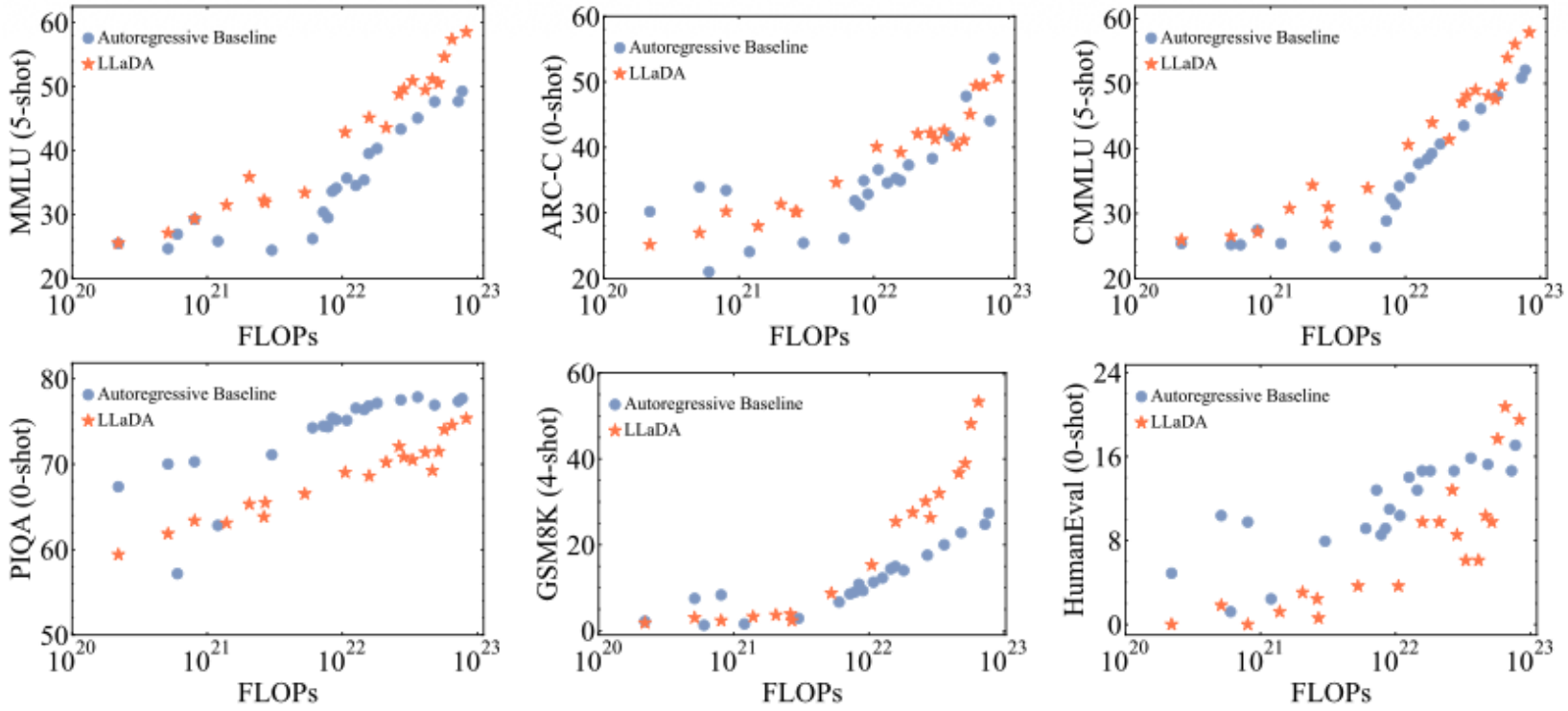


Figure 3: **Scalability of LLaDA.** We evaluate the performance of LLaDA and our ARM baselines trained on the same data across increasing pre-training computational FLOPs. LLaDA exhibits strong scalability, matching the overall performance of ARMs on six tasks.

What Did LLaDA Achieve?

Table 1: **Benchmark Results of Pre-trained LLMs.** * indicates that models are evaluated under the same protocol, detailed in Appendix B.6. Results indicated by † and ¶ are sourced from Yang et al. [25, 26] and Bi et al. [32] respectively. The numbers in parentheses represent the number of shots used for in-context learning. "-" indicates unknown data.

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B†	Qwen2.5 7B†	Mistral 7B†	Deepseek 7B¶
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	-	2T
General Tasks							
MMLU	**65.9** (5)	65.4 (5)	45.9 (5)	70.3 (5)	74.2 (5)	64.2 (5)	48.2 (5)
BBH	49.7 (3)	**62.1** (3)	39.4 (3)	62.3 (3)	70.4 (3)	56.1 (3)	39.5 (3)
ARC-C	45.9 (0)	**53.1** (0)	46.3 (0)	60.6 (25)	63.7 (25)	60.0 (25)	48.1 (0)
Hellaswag	70.5 (0)	**79.1** (0)	76.0 (0)	80.7 (10)	80.2 (10)	83.3 (10)	75.4 (0)
TruthfulQA	**46.1** (0)	44.0 (0)	39.0 (0)	54.2 (0)	56.4 (0)	42.2 (0)	-
WinoGrande	74.8 (5)	**77.3** (5)	72.5 (5)	77.0 (5)	75.9 (5)	78.4 (5)	70.5 (0)
PIQA	73.6 (0)	**80.6** (0)	79.1 (0)	-	-	-	79.2 (0)
Mathematics & Science							
GSM8K	**70.3** (4)	48.7 (4)	13.1 (4)	80.2 (4)	85.4 (4)	36.2 (4)	17.4 (8)
Math	**31.4** (4)	16.0 (4)	4.3 (4)	43.5 (4)	49.8 (4)	10.2 (4)	6.0 (4)
GPQA	25.2 (5)	**25.9** (5)	25.7 (5)	30.8 (5)	36.4 (5)	24.7 (5)	-
Code							
HumanEval	**35.4** (0)	34.8 (0)	12.8 (0)	51.2 (0)	57.9 (0)	29.3 (0)	26.2 (0)
HumanEval-FIM	**73.8** (2)	73.3 (2)	26.9 (2)	-	-	-	-
MBPP	40.0 (4)	**48.8** (4)	23.2 (4)	64.2 (0)	74.9 (0)	51.1 (0)	39.0 (3)
Chinese							
CMMLU	**69.9** (5)	50.7 (5)	32.5 (5)	83.9 (5)	-	-	47.2 (5)
C-Eval	**70.5** (5)	51.7 (5)	34.0 (5)	83.2 (5)	-	-	45.0 (5)

What Did LLaDA Achieve?

Table 2: **Benchmark Results of Post-trained LLMs.** LLaDA only employs an SFT procedure, while other models have extra reinforcement learning (RL) alignment. * indicates models are evaluated under the same protocol, detailed in Appendix B.6. Results indicated by † and ¶ are sourced from Yang et al. [26] and Bi et al. [32] respectively. The numbers in parentheses represent the number of shots used for in-context learning. "-" indicates unknown data.

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B†	Qwen2.5 7B†	Gemma2 9B†	Deepseek 7B¶
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	8T	2T
Post-training	SFT	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL	SFT+RL
Alignment pairs	4.5M	-	-	0.5M + -	1M + 0.15M	-	1.5M + -
General Tasks							
MMLU	65.5 (5)	**68.4** (5)	44.1 (5)	-	-	-	49.4 (0)
MMLU-pro	37.0 (0)	**41.9** (0)	4.6 (0)	44.1 (5)	56.3 (5)	52.1 (5)	-
Hellaswag	74.6 (0)	**75.5** (0)	51.5 (0)	-	-	-	68.5 (-)
ARC-C	**88.5** (0)	82.4 (0)	57.3 (0)	-	-	-	49.4 (-)
Mathematics & Science							
GSM8K	69.4 (4)	**78.3** (4)	29.0 (4)	85.7 (0)	91.6 (0)	76.7 (0)	63.0 (0)
Math	**31.9** (0)	29.6 (0)	3.8 (0)	52.9 (0)	75.5 (0)	44.3 (0)	15.8 (0)
GPQA	**33.3** (5)	31.9 (5)	28.4 (5)	34.3 (0)	36.4 (0)	32.8 (0)	-
Code							
HumanEval	49.4 (0)	**59.8** (0)	16.5 (0)	79.9 (0)	84.8 (0)	68.9 (0)	48.2 (-)
MBPP	41.0 (4)	**57.6** (4)	20.6 (4)	67.2 (0)	79.2 (0)	74.9 (0)	35.2 (-)

Takeaways

	Key Takeaway
Modeling	From discrete scores to masked-token prediction
Scaling	Competitive LLMs without autoregressive training
Generation	Flexible sampling with a single trained model
Challenges	Iterative decoding, KV caching, and inference efficiency

Autoregression is a choice, not a necessity.

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